

Docking Universal - Docking Study Report

Report generator version 0.6.0

Target: 1HVR | Ligand: XK2

1. Configured docking protocol

This table identifies the reusable target-matched protocol, records whether it was approved during this run or reused from an earlier run, and lists its locked docking settings.

Approved docking protocol	Selected value
Protocol file name	1HVR_XK2_20260821_135438.duprotocol
Approval provenance	Approved during this run
Control approval date (UTC)	2026-08-21T17:54:38.734634+00:00
Prepared receptor	1HVR.pdbqt
Locked docking box / pocket	1HVR_pocket1.conf
Engine	vina
Tier	repeatability
Exhaustiveness	16
Modes per job	15
Conformers per state	3
Independent seeds	5
Charge model	gasteiger
pH	7.4
Conformer force field	mmff94
Tautomers enumerated	True
Conformer RMSD pruning	0.75 A

2. Experimental control redocking results: XK2

This retrospective control tests whether the protocol reproducibly recovers the experimental pose of XK2. PASS requires sampling, ranking, and independent-seed criteria to pass. It supports use of the selected protocol for this target, but does not establish affinity or prospective pose accuracy.

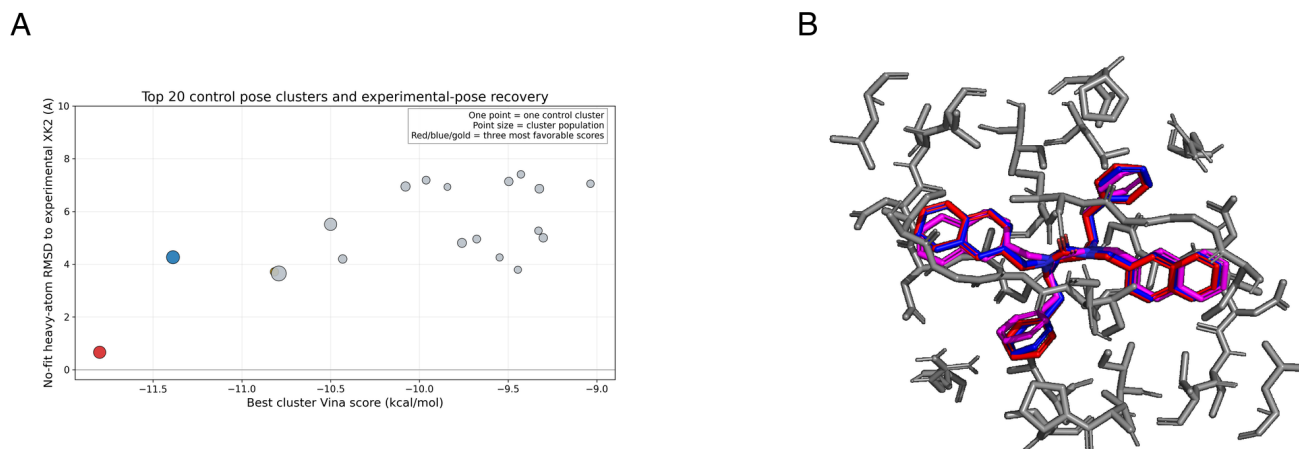


Figure 1. Retrospective control performance and pose recovery. (A) Control-cluster Vina score versus symmetry-aware, no-fit heavy-atom RMSD to experimental XK2 in the receptor coordinate frame. (B) Experimental ligand (magenta), lowest-energy pose (red), and lowest-RMSD pose (blue) superimposed in that frame; receptor residues within 5 Å of the displayed ligands are gray.

Control interaction diagrams

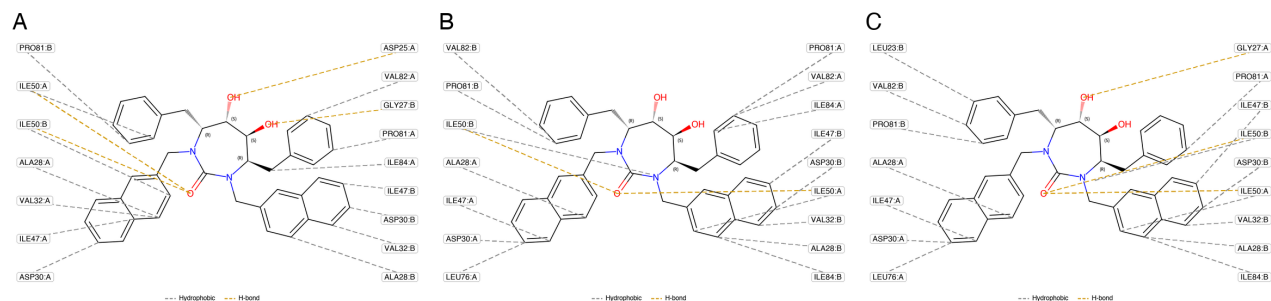


Figure 2. SDF-aware PLIP interaction diagrams for control ligand XK2. A is the experimental reference pose, B is the globally lowest-energy redocked pose, and C is the lowest-RMSD redocked pose. Ligand chemistry comes from the retained SDF files; interaction calls come from the retained PLIP XML.

3. New-ligand docking results

Summary of docking results

Summary of docking results	Recorded result
Ligand docked	Rilpivirine
Completion status	COMPLETED
Best retained Vina score	-9.21 kcal/mol
Top-ranked cluster	2
Top-cluster population	38
Independent-seed support	5
Conformer support	3
Distinct retained clusters	13
Selected representatives	3

Target: 1HVR | Ligand: Rilpivirine

Docking used the target-matched protocol established and approved by the control reported above. More favorable Vina scores are more negative; RMSD and cluster population are separate measures.

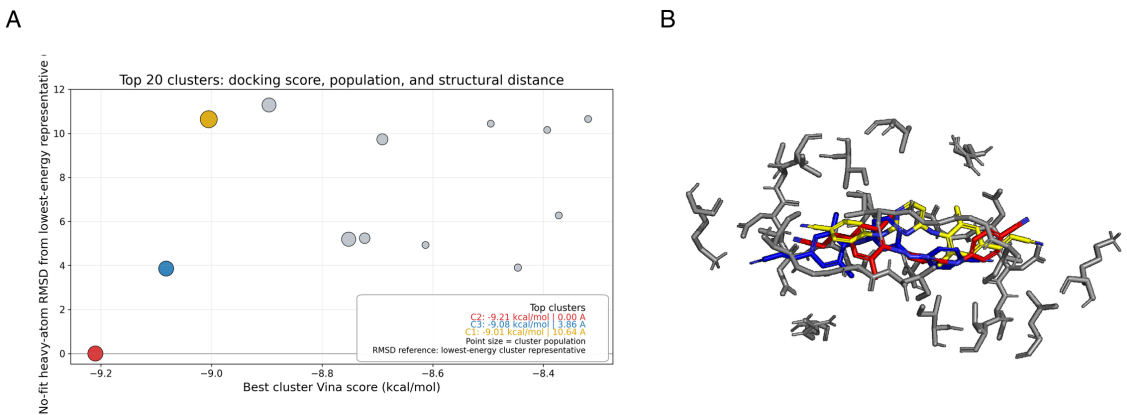


Figure 3. Docking pose-cluster analysis for Rilpivirine. (A) Docking score versus symmetry-aware, no-fit heavy-atom RMSD from the lowest-energy cluster representative in the receptor coordinate frame; point size denotes cluster population. (B) Representative structures from the highlighted clusters, using matching cluster colors.

Clusters represented in the docking plot

Rank	Cluster	Vina score	RMSD (Å)	Population
1	Cluster 2	-9.21	0.000000	38
2	Cluster 3	-9.082	3.858486	37
3	Cluster 1	-9.005	10.640349	50
4	Cluster 4	-8.896	11.288869	31
5	Cluster 5	-8.752	5.192835	32
6	Cluster 8	-8.723	5.238141	14
7	Cluster 7	-8.691	9.729086	16
8	Cluster 6	-8.613	4.929183	1
9	Cluster 10	-8.495	10.439236	1
10	Cluster 12	-8.446	3.902452	2
11	Cluster 13	-8.393	10.154944	1
12	Cluster 9	-8.372	6.272070	1
13	Cluster 11	-8.319	10.650684	1

Energy-ranked 3D cluster representatives

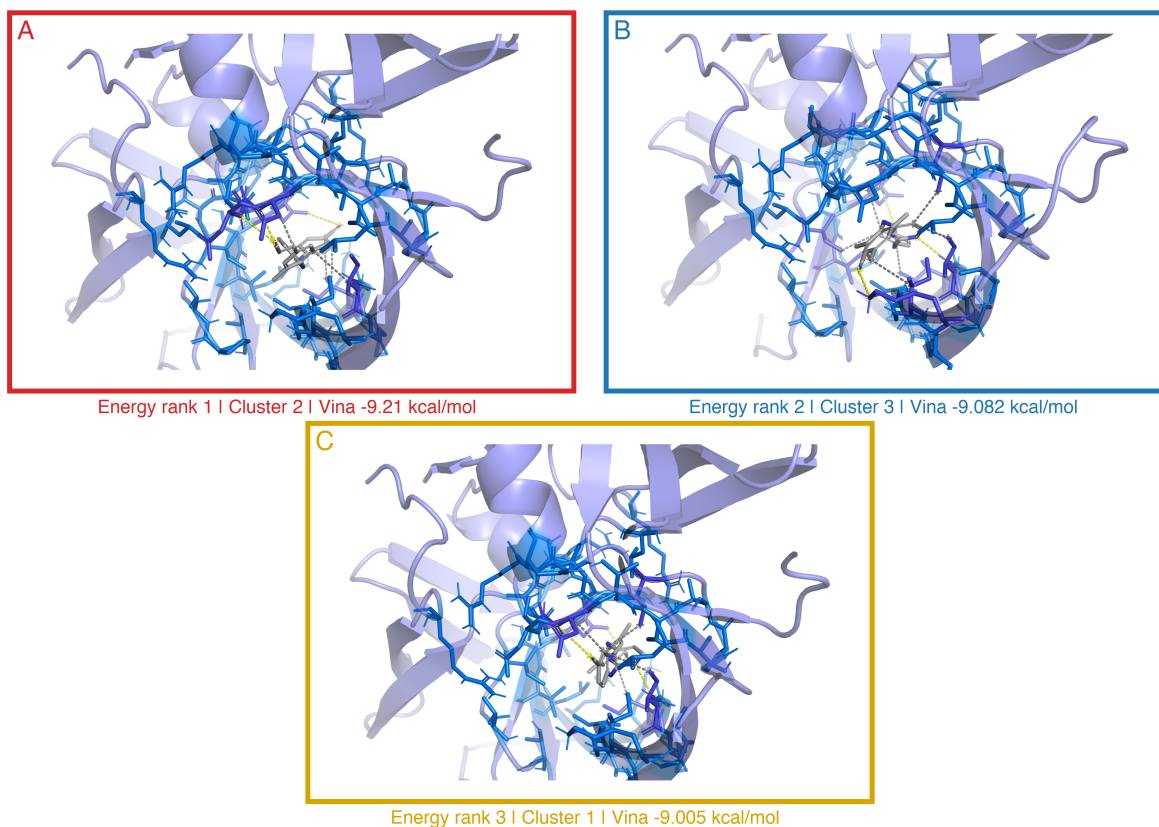


Figure 4. Three-dimensional interaction snapshots for Rilpivirine. Shown are 3 energy-ranked distinct cluster representatives, ordered by Vina score. Panels A, B, and C show energy ranks 1, 2, and 3, respectively. Red, blue, and gold match the highlighted clusters in Figure 3. These views support structural inspection; docking score rank does not establish pose correctness.

Selected 2D pose interaction diagrams

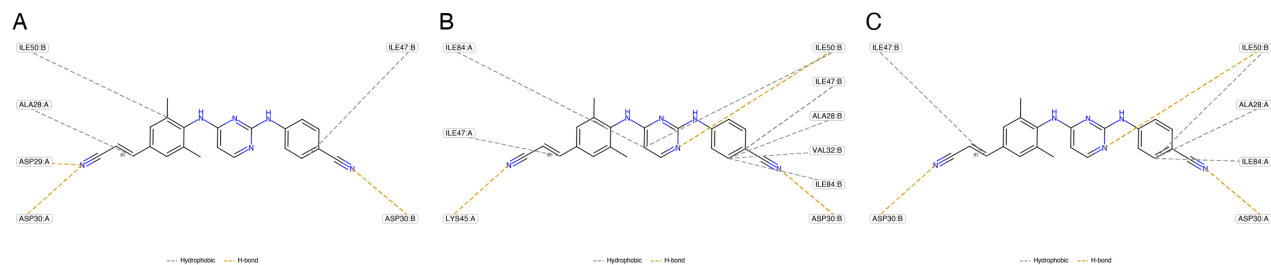


Figure 5. SDF-aware PLIP interaction diagrams for Rilpivirine. A, B, and C are the red, blue, and gold energy-ranked cluster representatives shown above. Ligand chemistry comes from each retained SDF; interaction calls come from the retained PLIP report.xml.

Interpretation and limitations

Docking scores are ranking estimates, not measured binding free energies. Rigid-receptor docking does not model induced fit. Cluster population and seed support describe computational convergence, not biological correctness. Protonation, tautomer, receptor preparation, and box choices can affect results. Experimental validation remains necessary.

4. Reproducibility, software, and references

Receptor preparation path: strict Meeko succeeded; PDBFixer was not needed. AutoDock Vina produced docking scores and poses. PLIP supplied rule-based protein-ligand interaction calls, with retained PLIP XML as the authoritative interaction record. RDKit supplied symmetry-aware heavy-atom RMSD calculations and Butina clustering at a 2.0 Å cutoff. PyMOL produced the molecular panels, and ReportLab assembled this PDF.

CCD/MODRES audit: A:67 CSO → CYS (resolved by an available Meeko template); B:67 CSO → CYS (resolved by an available Meeko template).

Control-to-new-run software check

Checked: no software version differences detected.

Software	Control	New run	Result
Docking Universal	0.6.0	0.6.0	SAME
Python	3.9.23	3.9.23	SAME
RDKit	2023.9.6	2023.9.6	SAME
MolScrub	0.2.2	0.2.2	SAME
Meeko	0.7.1	0.7.1	SAME
AutoDock Vina	AutoDock Vina v1.2.7	AutoDock Vina v1.2.7	SAME

Additional software used for analysis and reporting

Result element	Software	Version used
Molecular conversion/PLIP backend	Open Babel	3.1.0
Interaction calls	PLIP	2.3.1
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0

Scientific and software references

1. Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. J Chem Inf Model. 2021;61:3891-3898. <https://doi.org/10.1021/acs.jcim.1c00203>
2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
3. Salentin S, Schreiber S, Haupt VJ, Adasme MF, Schroeder M. PLIP: fully automated protein-ligand interaction profiler. Nucleic Acids Res. 2015;43:W443-W447. <https://doi.org/10.1093/nar/gkv315>
4. Butina D. Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. J Chem Inf Comput Sci. 1999;39:747-750. <https://doi.org/10.1021/ci9803381>
5. O'Boyle NM, Banck M, James CA, Morley C, Vandermeersch T, Hutchison GR. Open Babel: An open chemical toolbox. J Cheminform. 2011;3:33. <https://doi.org/10.1186/1758-2946-3-33>
6. RDKit: Open-source cheminformatics software. <https://www.rdkit.org/>
7. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>